

TILLAMOOK, OR, USA

WMO: 720202

Lat: 45.417N

Lon: 123.817W

Elev: 11

StdP: 101.19

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00118

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.4	-1.2	-10.0	1.6	2.3	-6.1	2.3	4.1	12.7	12.7	11.1	9.9	1.5	140	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.9	27.1	16.9	23.7	16.7	22.1	16.4	18.6	23.4	17.8	21.4	17.1	20.2	3.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	18.7	16.3	11.6	18.4	15.9	11.3	18.2	52.6	23.8	50.1	21.9	48.2	20.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.2	DB	-5.4	34.0	2.1	2.2	-7.0	35.6	-8.2	36.9	-9.4	38.2	-11.0	39.8	(4)
(5)			WB	-5.8	21.2	2.1	1.4	-7.3	22.2	-8.6	22.9	-9.8	23.7	-11.4	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	7.0	6.9	8.0	9.4	11.9	14.1	15.9	16.5	15.1	11.9	9.0	6.1	(6)	
	DBStd	4.40	2.86	2.96	2.52	2.32	2.39	2.14	1.71	1.80	2.34	2.60	3.10	3.36	(7)	
	HDD10.0	501	99	92	73	37	7	0	0	0	0	11	55	127	(8)	
	HDD18.3	2692	351	320	321	269	201	130	78	62	101	200	281	379	(9)	
	CDD10.0	866	6	5	11	18	66	122	183	201	153	69	24	7	(10)	
	CDD18.3	14	0	0	0	0	1	2	3	5	4	0	0	0	(11)	
	CDH23.3	312	0	0	0	8	39	26	43	79	104	11	0	0	(12)	
	CDH26.7	102	0	0	0	1	13	9	17	32	29	1	0	0	(13)	
(14) Wind	WSAvg	2.0	2.3	2.2	2.3	2.2	2.1	2.1	2.2	1.7	1.5	1.5	2.0	2.3	(14)	
(15) Precipitation	PrecAvg	2331	366	266	246	170	116	83	35	43	90	188	344	378	(15)	
	PrecMax	3117	694	627	469	368	215	192	136	166	278	360	645	614	(16)	
	PrecMin	1564	22	59	30	57	15	18	0	0	2	16	65	106	(17)	
	PrecStd	388	157	123	96	75	50	46	33	40	63	98	135	143	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	17.3	20.1	23.0	27.9	26.2	28.0	31.4	30.0	25.0	19.0	16.2	(19)	
		MCWB	10.6	11.1	11.5	14.4	15.9	18.7	18.5	18.3	16.3	14.5	14.1	13.2	(20)	
	2%	DB	13.9	14.9	16.2	18.8	22.2	22.1	22.8	24.9	27.0	21.0	17.0	14.0	(21)	
		MCWB	10.6	10.0	11.1	12.9	14.9	16.8	17.1	17.9	16.8	13.8	13.4	11.6	(22)	
	5%	DB	12.7	12.8	13.9	16.1	17.8	20.0	21.4	22.4	22.6	18.0	15.1	12.7	(23)	
		MCWB	10.7	10.1	10.1	11.3	13.8	15.5	16.7	17.1	16.5	14.3	13.0	11.0	(24)	
	10%	DB	12.1	12.0	12.5	13.8	16.3	18.0	20.2	21.1	20.2	17.1	13.9	12.0	(25)	
		MCWB	10.5	10.1	10.0	10.5	12.7	14.4	16.0	16.6	15.5	14.1	12.2	10.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.9	13.3	15.2	17.4	19.3	19.1	19.4	20.0	18.0	15.9	14.4	(27)	
		MCDB	14.0	15.9	16.7	21.8	24.5	25.4	24.7	25.3	24.1	19.5	16.9	15.8	(28)	
	2%	WB	12.3	11.8	12.0	13.3	15.3	17.2	17.9	18.4	18.3	16.5	14.5	12.6	(29)	
		MCDB	13.0	13.0	14.2	17.5	20.7	20.6	21.9	23.2	22.1	17.9	15.9	13.6	(30)	
	5%	WB	11.4	11.0	11.3	12.1	14.0	16.0	17.2	17.6	17.3	15.5	13.6	11.6	(31)	
		MCDB	12.2	12.2	13.2	14.4	17.9	18.9	20.3	21.1	20.5	17.2	15.1	12.5	(32)	
	10%	WB	10.3	10.1	10.5	11.3	13.1	15.1	16.5	16.9	16.3	14.6	12.5	10.3	(33)	
		MCDB	11.3	11.3	12.3	13.3	16.4	17.8	19.7	20.2	19.4	16.5	13.8	11.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.5	8.5	8.6	8.6	8.6	9.5	9.9	11.7	10.1	8.2	7.3	(35)	
		MCDBR	9.6	10.5	11.6	13.2	12.9	11.9	11.9	13.8	16.9	13.0	9.3	7.6	(36)	
	5% WB	MCWBR	6.4	6.8	7.3	8.1	7.7	6.9	6.8	7.5	8.9	7.9	6.3	5.6	(37)	
		MCWBR	6.7	7.7	9.4	10.6	11.3	10.1	10.2	11.3	12.4	8.7	7.3	6.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.322	0.337	0.366	0.378	0.364	0.349	0.357	0.348	0.336	0.330	0.328	(40)	
		taud	2.509	2.545	2.511	2.393	2.386	2.441	2.508	2.489	2.517	2.546	2.514	2.472	(41)	
	Ebn at Noon		785	868	898	894	891	902	913	893	874	836	772	736	(42)	
		Edh at Noon	68	78	91	112	116	111	102	100	90	76	67	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.20	2.05	3.03	4.36	5.45	5.96	6.52	5.58	4.32	2.56	1.40	0.99	(44)	
	RadStd		0.18	0.26	0.36	0.38	0.49	0.50	0.49	0.41	0.36	0.30	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+22

Nomenclature: See separate page